

Sakshi Kshirsagar

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Profile

Applied Machine Learning Engineer with hands-on experience building **end-to-end, deployed machine learning systems** using Python and Scikit-Learn. Strong background in **NLP, feature engineering, predictive modeling, and data pipelines**, with experience deploying ML applications to cloud platforms for real-time inference. Proven ability to translate business problems into **data-driven ML solutions**.

Projects

Content Virality Predictor (Deployed)

- **Python | Scikit-Learn | NLP | YouTube API | Render**
- Built an **end-to-end ML system** to predict early-stage content virality from YouTube metadata.
- Improved model **F1-score by 17%** through targeted NLP feature engineering.
- **Deployed** the model on Render with real-time inference via a web interface.
- Implemented a clean inference pipeline: preprocessing → prediction → output insights.

AI Chat Agent for Company Knowledge Base (RAG)

- **Python | NLP | Embeddings | RAG | OpenAI API**
- Designed a retrieval-augmented chatbot over **500+ pages of internal documents**.
- Built an embedding-based semantic search pipeline, improving answer relevance by **30%** over keyword search.
- Reduced dependency on manual support by enabling self-service information retrieval.

Credit Card Fraud Detection (Imbalanced Data)

- **Python | Logistic Regression | SMOTE | Classification**
- Built a fraud detection pipeline handling extreme class imbalance using **SMOTE**.
- Optimized models for **high recall**, minimizing business-critical false negatives.
- Evaluated using Precision, Recall, and F1-score for deployment reliability.

Skills & Abilities

- **Programming:** Python, SQL
- **Machine Learning:** Regression, Classification, Clustering, Model Evaluation, Feature Engineering
- **NLP & AI:** Natural Language Processing, LLMs, Embeddings, Retrieval-Augmented Generation (RAG)
- **Deployment & Tools:** GitHub, Render, REST APIs
- **Data & Databases:** Data Cleaning, Data Pipelines, MongoDB, MySQL
- **Visualization & BI:** Power BI, Tableau, Matplotlib, Seaborn
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-Learn, TensorFlow

Experience

Data Analyst Intern — Segelboot IT Solution Pvt. Ltd. Sept 2025 - Present

- Cleaned and transformed **50K+ rows** of multi-source university ranking data.
- Performed in-depth EDA on QS, THE, and ARWU datasets to identify key ranking drivers.
- Built analytical dashboards and reports to support leadership decision-making.
- Delivered complete analysis pipelines using Python, SQL, Pandas, and NumPy.

Education

- **MCA (AI/ML) — Ramdeobaba University, Nagpur — 2026**
- **B.Sc (Computer Science) — Janata Mahavidyalaya, Chandrapur — 2024**